DO NOW — RECALL

Lesson 081 · Merge Sort: Bottom-up Merge Tracing

Name: _____ Class: _____ Date: _____

1. What is a sorted run?

2. How does insertion sort work?

3. What is O(n log n) complexity?

4. Binary search sorted [1,3,5,7,9,11,13] for 11: show midpoint index sequence.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Contiguous segment already in ascending order.

2. Model answer

Inserts each next key into its sorted prefix.

3. Model answer

Work scaling with n times log n.

4. Model answer

mid3=7 then mid5=11; return index5.

By the end, you can…

• Merge sort divides the input into smaller runs; merging two already sorted runs repeatedly selects the smallest available front value.

• Bottom-up merge sort begins with runs of length one and doubles run width on every complete pass; it does not require recursive calls.

• Worst-case time O(n log n); standard array merging uses O(n) extra space. Include comparisons, not just the final sorted sequence.

Vocabulary — know these terms

merge sort — Sort by merging sorted runs.

run — Consecutive already sorted segment.

merge — Combine two sorted sequences.

auxiliary space — Extra memory beyond input.

NEW LEARNING

Read, annotate and explain

Observe

Merge sort divides the input into smaller runs; merging two already sorted runs repeatedly selects the smallest available front value.

Apply

This bottom-up variant starts with single-item runs and doubles run width; it needs no recursive calls. The top-down variant recursively splits to length 0/1, then merges on return; Lesson 186 connects the two.

Evaluate

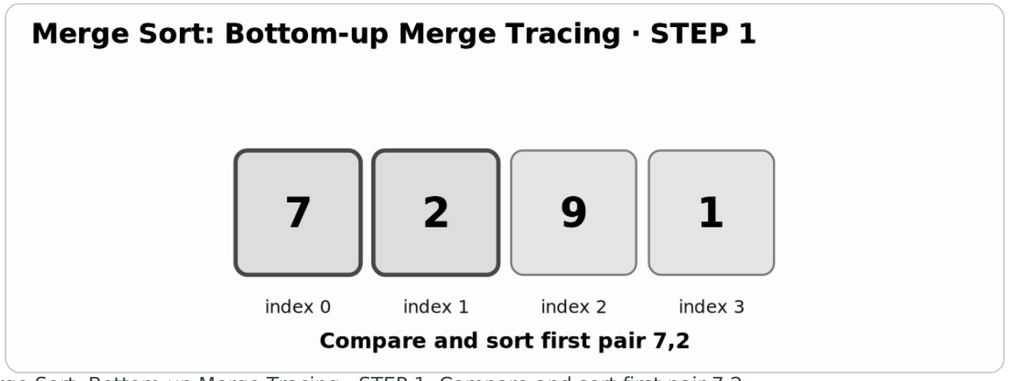
Worst-case time O(n log n); standard array merging uses O(n) extra space. Include comparisons, not just the final sorted sequence.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

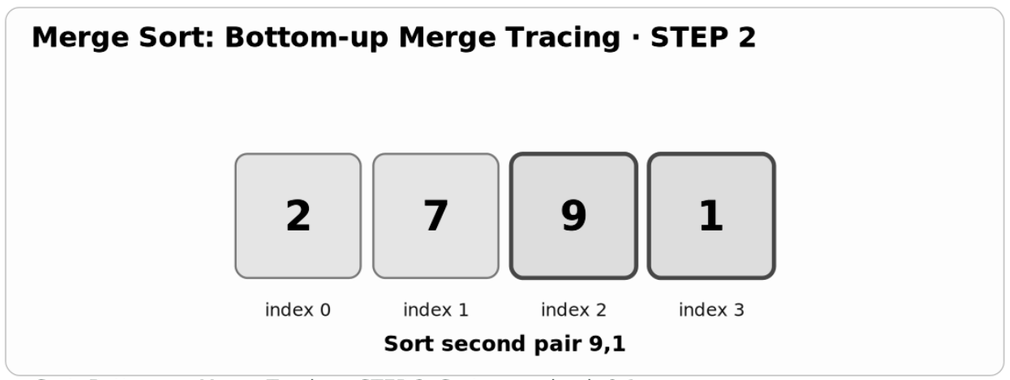
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Merge Sort: Bottom-up Merge Tracing · STEP 1; Compare and sort first pair 7,2

New Learning



Merge Sort: Bottom-up Merge Tracing · STEP 2; Sort second pair 9,1

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Bottom-up merge [5,2,7,4]; trace first two sorted pairs and final merge.

Pair sort [2,5,4,7]; merge sorted runs → [2,4,5,7].



WEDO · STATE; Merged runs produce ascending output

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

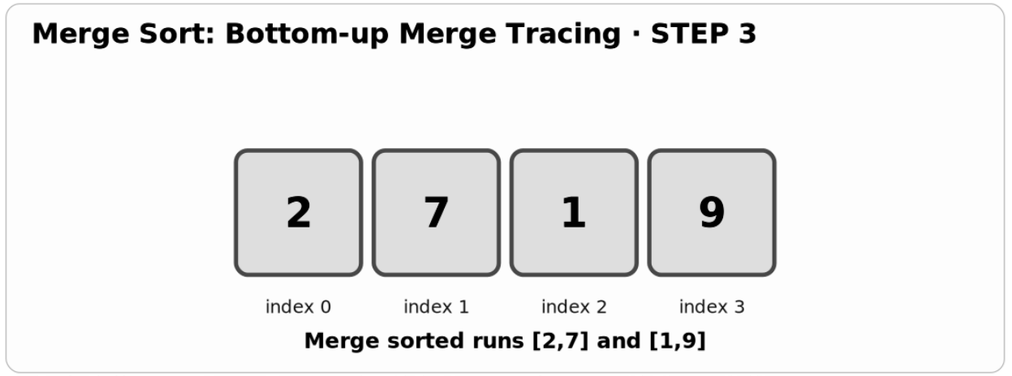
Guided 1 — CHECK

Method and result
Pair sort [2,5,4,7]; merge sorted runs → [2,4,5,7].

New Learning · original lesson diagrams

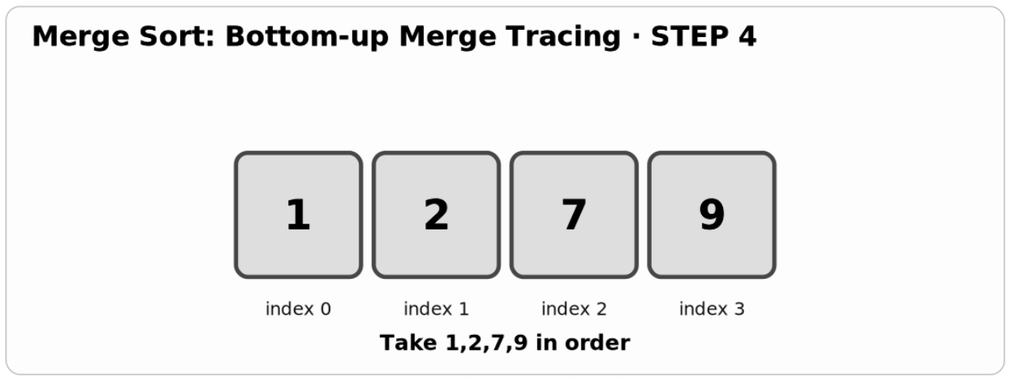
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Merge Sort: Bottom-up Merge Tracing · STEP 3; Merge sorted runs [2,7] and [1,9]

New Learning



Merge Sort: Bottom-up Merge Tracing · STEP 4; Take 1,2,7,9 in order

I DO — WORKED EXAMPLE

Follow the worked method

Bottom-up merge [8,3,6,1]; show pair results and full result.

Method and result

Pairs [3,8] and [1,6] → [3,8,1,6]; merge fronts → [1,3,6,8].

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Merge sort requires no additional array storage in its standard array implementation.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

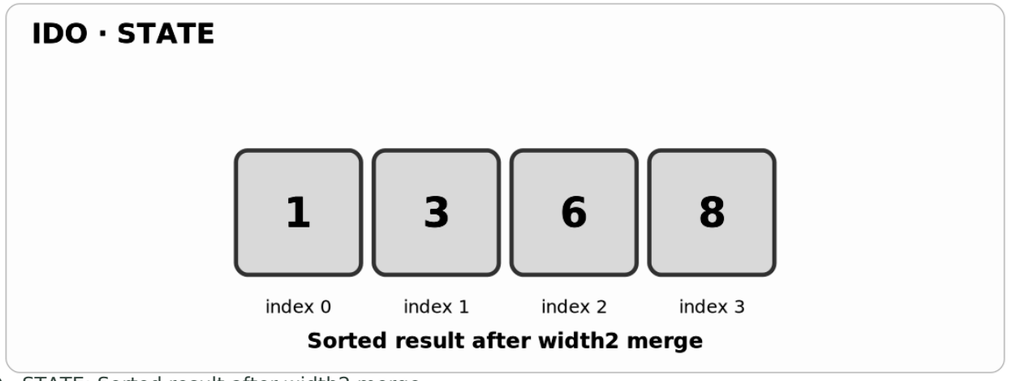
Compare bottom-up merge with insertion sort for large random data. Give worst time and auxiliary-space cost. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Bottom-up merge [8,3,6,1]; show pair results and full result.

Pairs [3,8] and [1,6] → [3,8,1,6]; merge fronts → [1,3,6,8].



IDO · STATE; Sorted result after width2 merge

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Bottom-up merge [5,2,7,4]; trace first two sorted pairs and final merge.
Source code / input:
Pair sort [2,5,4,7]; merge sorted runs → [2,4,5,7].

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Given [7,2,9,1], state array after width1 pair merges and final width2 merge. [6]

Question 2

Merge sorted runs [2,7] and [1,9] and list chosen values in order, with comparisons. [6]